

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12415225
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Zhou; Yang et al.

Handheld band saw

Abstract

A handheld band saw includes a housing assembly including a battery pack mounting portion for mounting a battery pack, a rotating wheel assembly at least partially disposed in the housing assembly and including a first rotating wheel and a second rotating wheel, where the first rotating wheel and the second rotating wheel are configured to be substantially symmetrical about a midplane, an electric motor disposed in the housing assembly and capable of driving one of the first rotating wheel and the second rotating wheel to rotate, a fan which is capable of generating at least one airflow in the housing assembly, and a control circuit board disposed in the housing assembly.

Inventors:	Zhou; Yang (Nanjing, CN), Gao; Hao (Nanjing, CN)
Applicant:	Nanjing Chervon Industry Co., Ltd. (Nanjing, CN)
Family ID:	1000008820885
Assignee:	Nanjing Chervon Industry Co., Ltd. (Nanjing, CN)
Appl. No.:	18/446239
Filed:	August 08, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240058879 A1	Feb. 22, 2024

Foreign Application Priority Data

CN	202210989063.9	Aug. 17, 2022
CN	20222172168.3	Aug. 17, 2022

Publication Classification

Int. Cl.: B23D53/12 (20060101); B23D55/02 (20060101); B23D55/06 (20060101); B23D55/08 (20060101)

U.S. Cl.:

CPC B23D53/12 (20130101); B23D55/02 (20130101); B23D55/065 (20130101); B23D55/082 (20130101);

Field of Classification Search

CPC: B23D (53/12); B23D (55/065); B23D (55/082); B23D (55/02); B23D (59/006); B23D (55/00); B25F (5/008)

USPC: 30/380; 30/371; 30/286; 30/320

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5363558	12/1993	Schroeder	30/500	B23D 57/0015
2007/0068011	12/2006	Shibata	30/380	B27B 13/16
2013/0333228	12/2012	Ota	30/273	B23D 47/12
2015/0026985	12/2014	Yamaguchi	30/123	B23D 45/16
2022/0263377	12/2021	Sasaki	N/A	H02K 7/116
2023/0007824	12/2022	Okuhira	N/A	B23D 53/12
2023/0112387	12/2022	Kachi	30/165	B27B 17/00

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
110744135	12/2019	CN	B23D 53/12
1980352	12/2007	EP	B23D 53/12
3778132	12/2020	EP	B01D 46/0005

Primary Examiner: Alie; Ghassem

Attorney, Agent or Firm: Greenberg Traurig, LLP

Background/Summary

RELATED APPLICATION INFORMATION

(1) This application claims the benefit under 35 U.S.C. § 119(a) of Chinese Patent Application No. CN 202210989063.9, filed on Aug. 17, 2022, and Chinese Patent Application No. CN 202222172168.3, filed on Aug. 17, 2022, which applications are incorporated herein by reference in their entireties.

BACKGROUND

(2) At present, handheld band saws are widely used and can be used for cutting all machinable metals including stainless steel, pipes, solid bodies, and the like. The handheld band saw has a strong thermal resistance, wear resistance, and vibration resistance and has a longer service life than similar products. Because the handheld band saw has advantages such as flexible and

convenient use, the handheld band saw has been widely used in industrial and household scenarios where various types of bars, pipes, and profiles are cut.

(3) A handheld band saw typically controls the running of an electric motor with a control circuit board. When the handheld band saw works, the temperature rise of the electric motor and the temperature rise of the control circuit board are relatively high, and the electric motor and the control circuit board are easily damaged, thereby affecting the use of the handheld band saw.

SUMMARY

(4) A handheld band saw includes: a housing assembly including a battery pack mounting portion for mounting a battery pack; a rotating wheel assembly at least partially disposed in the housing assembly and including a first rotating wheel and a second rotating wheel, where the first rotating wheel and the second rotating wheel are configured to be substantially symmetrical about a midplane; an electric motor disposed in the housing assembly and capable of driving one of the first rotating wheel and the second rotating wheel to rotate; a fan which is capable of generating at least one airflow in the housing assembly during rotation; and a control circuit board disposed in the housing assembly. The housing assembly is formed with an air inlet and an air outlet, and when the fan rotates, the airflow entering the housing assembly from the air inlet flows through at least the control circuit board and the electric motor.

(5) A handheld band saw includes: a housing assembly including a battery pack mounting portion for mounting a battery pack; a rotating wheel assembly at least partially disposed in the housing assembly and including a first rotating wheel and a second rotating wheel, where the first rotating wheel and the second rotating wheel are configured to be substantially symmetrical about a midplane; an electric motor disposed in the housing assembly and capable of driving one of the first rotating wheel and the second rotating wheel to rotate; a control circuit board disposed in the housing assembly; and a fan which is capable of generating at least one airflow flowing through the control circuit board in the housing assembly during rotation.

(6) The control circuit board and the electric motor are located on the same side of the midplane.

(7) A handheld band saw includes: a housing assembly; an electric motor disposed in the housing assembly; a fan capable of being driven by the electric motor and capable of generating at least one airflow in the housing assembly during rotation; and a control circuit board disposed in the housing assembly. The housing assembly is formed with an air inlet and an air outlet, and when the fan rotates, at least one airflow entering the housing assembly from the air inlet passes through at least the control circuit board and the electric motor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a handheld band saw in the present application;

(2) FIG. 2 is a plan view of the handheld band saw shown in FIG. 1;

(3) FIG. 3 is a perspective view of the handheld band saw shown in FIG. 1 from another angle of view;

(4) FIG. 4 is a perspective view of the handheld band saw shown in FIG. 1 from another angle of view;

(5) FIG. 5 is a sectional view of the handheld band saw shown in FIG. 4 along an A-A direction;

(6) FIG. 6 is a sectional view of the handheld band saw shown in FIG. 4 along a B-B direction;

(7) FIG. 7 is a schematic view showing part of the structures of the handheld band saw shown in FIG. 1;

(8) FIG. 8 is a schematic view showing part of the structures of the handheld band saw shown in FIG. 1;

(9) FIG. 9 is a schematic view of the handheld band saw shown in FIG. 1 with a shield housing

opened;

(10) FIG. 10 is an exploded view of locking assemblies in the handheld band saw shown in FIG. 9;

(11) FIG. 11 is a partial enlarged view of the handheld band saw shown in FIG. 8; and

(12) FIG. 12 is a perspective view of a stopper in a locking assembly shown in FIG. 9.

DETAILED DESCRIPTION

(13) FIG. 1 shows a power tool, which is specifically a handheld band saw 100. The handheld band saw may drive a functional element 51 to move. The functional element 51 is specifically a saw blade. In this manner, the handheld band saw 100 may drive the saw blade to cut pieces to be cut. The pieces to be cut are all machinable metals including stainless steel, pipes, solid bodies, and the like.

(14) To clearly illustrate the technical solutions of the present application, up, down, front, rear, left, and right as shown in FIG. 1 are defined.

(15) Referring to FIGS. 1 to 4, an example of the present disclosure provides a power tool, which is specifically the handheld band saw 100. The handheld band saw 100 of the present disclosure includes a main body 10 and an energy source 11. The main body 10 includes a housing assembly 20, a power assembly 30, a rotating wheel assembly 40, a cutting assembly 50, a handle assembly 60, a switch assembly 62, a heat dissipation device 70, and a shield assembly 80. The energy source 11 may be specifically a direct current or an alternating current. In this example, the energy source 11 is specifically a battery pack 12.

(16) The housing assembly 20 is a main part constituting the appearance of the handheld band saw 100. The housing assembly 20 has a first space 21 for at least partially accommodating various components in the handheld band saw 100. That is, the power assembly 30, the rotating wheel assembly 40, the cutting assembly 50, the switch assembly 62, and the heat dissipation device 70 are at least partially disposed in the first space 21.

(17) The power assembly 30 is used for providing power for the handheld band saw 100. The power tool includes an electric motor 31 including an electric motor shaft 32 rotatable about an electric motor axis 101. The electric motor shaft 32 has a first distance D1 along the direction of the electric motor axis 101. It is to be noted that in this example, the extension direction of the electric motor axis 101 substantially coincides with an up and down direction.

(18) The rotating wheel assembly 40 is used for transmitting the power from the power assembly 30 to the cutting assembly 50. The rotating wheel assembly 40 includes a first rotating wheel 41 and a second rotating wheel 42, and the electric motor 31 can drive the first rotating wheel 41 and/or the second rotating wheel 42 to move. In the present application, the electric motor 31 drives the first rotating wheel 41 to rotate about a first axis. The first rotating wheel 41 and the second rotating wheel 42 are spaced to form a cutting region 26 and are configured to be substantially symmetrical about a midplane P1. The first axis is parallel to or coincides with the electric motor axis 101.

(19) In fact, the housing assembly 20 includes a main housing 22, a first rotating wheel housing 23, a second rotating wheel housing 24, and an electric motor housing 25. The first rotating wheel 41 and the second rotating wheel 42 are connected to the main housing 22. The first rotating wheel housing 23 is formed with a first open space 231, and the second rotating wheel housing 24 is formed with a second open space 241. The first rotating wheel 41 is substantially accommodated in the first open space 231, and the second rotating wheel 42 is substantially accommodated in the second open space 241, that is to say, the first rotating wheel housing 23 substantially accommodates the first rotating wheel 41, and the second rotating wheel housing 24 substantially accommodates the second rotating wheel 42. The first rotating wheel housing 23, the second rotating wheel housing, and the electric motor housing 25 are fixedly connected to the main housing 22. The midplane P1 is a virtual plane. The main housing 22, the first rotating wheel housing 23, and the electric motor housing 25 may be made of different materials, and the main housing 22 may be made of aluminum alloy. The cutting assembly 50 is used for outputting the

power and includes the functional element **51**, that is, the saw blade. The functional element **51** connects the first rotating wheel **41** to the second rotating wheel **42** and drives the second rotating wheel **42** to move synchronously with the first rotating wheel **41**. When the main body **10** moves, the functional element **51** moves with the first rotating wheel **41**. During the movement, the functional element **51** is at least partially located in the cutting region **26**. The functional element **51** located in the cutting region **26** approaches a piece to be cut to cut it. The cutting assembly **50** further includes two pairs of guide assemblies **52** disposed opposite to each other and configured to be symmetrical about the midplane **P1**. The guide assemblies **52** are disposed near the cutting region **26**. In addition, with the guide assemblies **52**, part of the saw blade located in the cutting region **26** performs cutting along a first cutting direction **103**.

(20) The handle assembly **60** includes a main handle **61**. The housing assembly **20** is connected to or formed with the main handle **61**. The housing assembly **20** includes the main handle **61** including a right end **614**, a left end **615**, and a handle portion **611** for a user to hold. The handle portion **611** is configured to be substantially symmetrical about a handle midplane **P2**. The handle midplane **P2** is the virtual plane. The main handle **61** includes a first handle housing portion **612** and a second handle housing portion **613**. The first handle housing portion **612** and the second handle housing portion **613** are detachably assembled to form a hollow accommodating space **65**. In the present application, the main handle **61** is part of the housing assembly **20**.

(21) The switch assembly **62** is used for controlling the electric motor **31** to be turned on and off. The switch assembly **62** is coupled to a control circuit board **64** for connecting the switch assembly **62** to the electric motor **31**. The switch assembly **62** is mounted on the handle portion **611**, that is, the switch assembly **62** and the control circuit board **64** are at least partially disposed in the accommodating space **65**. With this configuration, the user can conveniently trigger the switch assembly **62** when holding the handle portion **611**.

(22) As shown in FIGS. **1** to **4**, the housing assembly **20** is connected to or formed with a battery pack mounting portion **27** for mounting the battery pack **12**. The battery pack mounting portion **27** is configured to be adjacent to the main handle **61**, and when the battery pack **12** is mounted on the mounting base of the battery pack **12**, the handle midplane **P2** passes through the battery pack **12**. Along a left and right direction, the battery pack mounting portion **27** and the electric motor **31** are disposed on two sides of the midplane **P1**. The battery pack **12** and the electric motor **31** are heavier than other components of the handheld band saw **100**. The battery pack **12** and the electric motor **31** are disposed at two ends of the handheld band saw **100** separately so that the overall weight of the handheld band saw **100** can be relatively balanced when the battery pack **12** is mounted on the main body **10**. The case does not occur where the handheld band saw **100** is relatively heavy on one side, resulting in that the whole handheld band saw **100** is oblique to the side when being held and performing the cutting. Thus, the operation of the user is facilitated.

(23) The heat dissipation device **70** is used for dissipating the heat of the electric motor **31**. The heat dissipation device **70** includes a fan **71**. The fan **71** can be driven by the electric motor shaft **32** to rotate. When the fan **71** rotates, at least one airflow is formed, and the airflow enters the inside of the housing assembly **20** from an air inlet and then flows out from an air outlet. On a flow path of the airflow, the airflow passes through at least the electric motor **31** and the control circuit board **64**, thereby dissipating the heat of the control circuit board **64** and the heat of the electric motor **31**.

(24) As an example, the housing assembly is formed with multiple air inlets. An airflow is formed and enters the housing assembly from each of the air inlets, that is to say, at least two airflows are formed when the fan rotates. One airflow flows through the electric motor and is used for dissipating the heat of the electric motor, the other airflow flows through the control circuit board and is used for dissipating the heat of the control circuit board, and finally the two airflows combine to flow out from the air outlet. The two airflows may flow out from multiple air outlets separately.

(25) As another example, the housing assembly is formed with multiple air inlets. An airflow is

formed and enters the housing assembly from each of the air inlets, that is to say, at least two airflows are formed when the fan rotates. One airflow flows through the electric motor and is used for dissipating the heat of the electric motor, the other airflow flows through the control circuit board and the electric motor and is used for dissipating the heat of the control circuit board and the heat of the electric motor, and finally the two airflows combine to flow out from the air outlet. The two airflows may flow out from multiple air outlets separately.

(26) As another example, the housing assembly is formed with one or more air inlets. An airflow is formed and enters the housing assembly from each air inlet, that is to say, at least one airflow is formed when the fan rotates. The one or more airflows each flow through the electric motor and the control circuit board and finally flow out from the air outlet. This example is described in detail below.

(27) When the fan **71** rotates, an airflow A is formed. The airflow A enters the inside of the housing assembly **20** from an air inlet **72** and flows out from an air outlet **73**. On a flow path of the airflow A, the airflow A passes through at least the electric motor **31** and the control circuit board **64** to dissipate the heat of the control circuit board **64** and the heat of the electric motor **31**.

(28) When the fan **71** rotates, the airflow A enters the housing assembly **20** from the air inlet **72**, and on the flow path of the airflow A, the airflow A passes through the control circuit board **64**, then flows to the electric motor **31**, and finally flows out from the air outlet **73**. When the handheld band saw **100** runs, the electric motor **31** rotates at a high speed and generates a large amount of heat. That is, the heat generated by the electric motor **31** is greater than the heat generated by the control circuit board **64**. That is to say, the temperature of the airflow A flowing through the electric motor **31** is bound to be higher than the temperature of the airflow A flowing through the control circuit board **64**. With the preceding configuration, the airflow A dissipates the heat of the control circuit board **64** and the heat of the electric motor **31** better so that it is ensured that the control circuit board **64** and the electric motor **31** run normally. In some possible examples, the air inlet **72** and the air outlet **73** are each disposed on one side of the midplane P1. With this configuration, the heat dissipation effect of the handheld band saw **100** can be ensured, and the dissipation structure is simple.

(29) The main handle **61** is provided with the inlet of the airflow A, and the electric motor housing **25** is provided with the outlet of the airflow A. The fan **71** may be a centrifugal fan **71**. When the fan **71** rotates, the centrifugal fan **71** generates negative pressure, and the airflow A is drawn into the accommodating space **65** of the main handle **61** from the inlet of the airflow A, then flows into the interior space **251** of the electric motor housing **25**, and is finally discharged from the air outlet **73** of the electric motor housing **25**, thereby performing the heat dissipation on the electric motor **31** and the control circuit board **64** and prolonging the life of the electric motor **31** and the life of the control circuit board **64**. In some possible examples, the air inlet **72**, the air outlet **73**, and the control circuit board **64** are each disposed on one side of the central plane P1, and the battery pack mounting portion **27** is disposed on the other side of the central plane P1. With this configuration, the heat of the control circuit board **64** can be better dissipated so that the heat dissipation effect thereof is better.

(30) The control circuit board **64** is disposed in the accommodating space **65**. Specifically, the control circuit board **64** is disposed on the side of the midplane P1 facing the electric motor **31**. That is to say, along the left and right direction, the control circuit board **64** is disposed on the left side of the midplane P1, the electric motor **31** is disposed on the left side of the midplane P1, and the battery pack mounting portion **27** is disposed on the right side of the midplane P1. That is, the control circuit board **64** is configured to be adjacent to the electric motor **31**. The accommodating space **65** where the control circuit board **64** is disposed communicates with the interior space **251** where the electric motor **31** is disposed such that a flow channel of the airflow A is formed. The electric motor housing **25** is formed with a connection port **252**. The airflow A entering the accommodating space **65** directly enters the interior space **251** from the connection port **252**. The

electric motor housing **25** includes a first electric motor housing **253** and a second electric motor housing **254**. The first electric motor housing **253** and the second electric motor housing **254** are each fixedly connected to the main housing **22** and are combined to form the connection port **252** communicating with the accommodating space **65**. The connection port **252** is at least partially located in the accommodating space **65**. The first handle housing portion **612** is formed with a through hole via which the electric motor housing **25** can at least partially extend into the accommodating space **65**. The through hole communicates with the connection port **252**. The interior space **251** and the accommodating space **65** may communicate with each other via a third through space, that is, the interior space **251** and the accommodating space **65** may communicate with each other indirectly. The control circuit board **64** is disposed at the left end of the main handle **61** and configured to be adjacent to the electric motor **31**. On the flow path of the airflow A, the airflow A needs to flow through the control circuit board **64**, the electric motor **31**, and the like. To provide the preceding heat dissipation path A, the air inlet **72** is configured to be adjacent to the control circuit board **64**, or the control circuit board **64** and the electric motor **31** are disposed between the air inlet **72** and the air outlet **73**. If the control circuit board **64** is disposed on the right side of the midplane **P1**, not only is a space needed for arranging the control circuit board **64**, but also an enough space needs to be retained in the main body for disposing the heat dissipation channel. In addition, to make the structure of the main body reasonable, the heat dissipation channel is configured to be narrow and long under this condition. Therefore, a poor heat dissipation effect of the main body is caused. The air inlet **72** is disposed at the left end of the main handle **61**. The air inlet **72** and the air outlet **73** are configured to be adjacent to each other. That is to say, the distance of the heat dissipation channel between the air inlet **72** and the air outlet **73** is relatively short. Thus, the air duct between the air inlet **72** and the air outlet **73** can be configured to be relatively spacious, thereby facilitating sufficient heat dissipation.

(31) Along the extension direction of the electric motor axis **101**, the fan **71** is disposed at the upper end of the electric motor **31**, and at least 90% of the connection port **252** is configured to be located at the lower end of the electric motor **31**. With this configuration, the airflow A entering from the connection port **252** can sufficiently flow through the electric motor **31** and then is discharged from the air outlet **73**.

(32) As shown in FIGS. 5, 8, and 11, the control circuit board **64** is disposed in the left end of the main handle **61**, and specifically, the control circuit board is configured to be oblique in the accommodating space **65**. The heat dissipation device **70** further includes a flow guide structure **74** fixedly disposed in the accommodating space **65** and disposed between the air inlet **72** and the control circuit board **64**. The control circuit board **64** is configured to be oblique in the accommodating space **65** and forms a space with the left end of the main handle **61**, and the flow guide structure **74** is disposed in the space. The flow guide structure **74** is used for guiding the airflow A to flow to the control circuit board **64**. The flow guide structure **74** is at least partially disposed between the air inlet **72** and the connection port **252**. The flow guide structure **74** includes a first guide portion **741** and a second guide portion **742**. The first guide portion **741** and the second guide portion **742** are each connected to the first handle housing portion **612** and the second handle housing portion **613**. The first guide portion **741** is partially connected to the second guide portion **742**. The first guide portion **741**, the second guide portion **742**, the first handle housing portion **612**, and the second handle housing portion **613** are connected to form a stopper structure for restricting the airflow A entering from the air inlet **72** from directly entering the interior space **251**. The first guide portion **741** extends substantially along a first direction **102**, where the first direction **102** intersects the direction of the electric motor axis **101** obliquely. The airflow A entering from the air inlet **72** is guided by the first guide portion **741** to flow upward along the first direction **102**, that is, the airflow A is guided to sufficiently flow to the control circuit board **64** to sufficiently dissipate the heat of the control circuit board **64**.

(33) The shield assembly **80** is connected to the housing assembly **20** and used for substantially

covering the first rotating wheel **41** and the second rotating wheel **42**. The shield assembly **80** is movably connected to the housing assembly **20** and has at least a first state and a second state during the movement of the shield assembly **80**. When the shield assembly **80** is in the first state, the shield assembly **80** is relatively fixed to the housing assembly **20** and can substantially cover the rotating wheel assembly **40**, which may be understood as the case where the shield assembly **80** is locked relative to the housing assembly **20**. When the shield assembly **80** is in the second state, the shield assembly **80** is relatively movable relative to the housing assembly **20**, which may be understood as the case where the shield assembly **80** is in a released state relative to the housing assembly **20**.

(34) The handheld band saw **100** further includes a locking device **81** movably connected to the housing assembly **20**. The locking device **81** is used for locking and releasing the shield assembly **80** to cause the shield assembly **80** to switch between a first state and a second state. The locking device **81** includes at least one group of locking assemblies **82**.

(35) A locking assembly **82** includes a driving member **821**, a stopper **822**, a biasing member **823**, and an abutting member **824**. The driving member **821** has a retainable open position and a retainable closed position. The driving member **821** may be driven by the user to the open position or the closed position. When the driving member **821** is at the open position, the shield assembly **80** is in the second state. In this case, the user can open the shield assembly **80**, replace the saw blade, clean chips, or the like. When the driving member **821** is at the closed position, the shield assembly **80** may be in the first state or may remain in the second state. When the shield assembly **80** is in the first state, the shield assembly **80** and the housing assembly **20** remain relatively fixed to each other. In this case, the user can start the electric motor **31** for cutting. When the driving assembly is at the closed position and the shield assembly **80** is in the second state, the shield assembly **80** is always in the moving state. The stopper **822** cooperates with the driving member **821** so that the driving member **821** can be retained at the open or closed position. The biasing member **823** always provides a driving force for the driving member **821** so that the driving member **821** is always remained at a current position without being driven by an external force. The abutting member **824** abuts against the driving member **821**. With this configuration, the user only needs to drive the driving member **821** such that the locking assembly **82** can be opened or closed, so as to lock or release the shield assembly **80**. In addition, the driving member **821** is always at the current position with being driven by the external force. That is to say, the user needs to operate, in only one step, the driving member **821** to release or lock the shield assembly **80**, thereby facilitating the operation of the user.

(36) As shown in FIGS. **9**, **10**, and **12**, the stopper **822** is fixedly connected to the main housing **22**, and in this example, the stopper **822** is fixedly connected to the main housing **22** by a screw. The stopper **822** is formed with slide grooves **825** and a movement space **826**. The slide grooves **825** protrude from the movement space **826**, that is, the slide grooves **825** are at least partially disposed outside the movement space **826**. The biasing member **823** is at least partially disposed in the movement space **826**. One end of the biasing member **823** abuts against the main housing **22**, and the other end of the biasing member **823** abuts against the abutting member **824**. When the abutting member **824** is forced to move toward the main housing **22**, the biasing member **823** is forced to be compressed in the movement space **826**. When the external force applied to the abutting member **824** is removed, the biasing member **823** rebounds, and the other end of the biasing member **823** continues abutting against the abutting member **824**. The abutting member **824** is at least partially disposed in the movement space **826**. One end of the abutting member **824** abuts against the biasing member **823** and the other end the abutting member **824** abuts against the driving member **821**, which may be understood as the case where the driving force of the biasing member **823** is transmitted to the driving member **821** through the abutting member **824**. The driving member is partially disposed in the slide grooves **825**, and the driving member **821** is movable relative to the slide grooves **825**.

(37) The driving member **821** cooperates with the stopper **822** so that the driving member **821** switches between the open position and the closed position in a retainable manner. The driving member **821** includes a first contact surface **827** and a second contact surface **828** which are not in the same plane. When the driving member **821** is at the open position, the first contact surface **827** is in contact with the abutting member **824**, and when the driving member **821** is at the closed position, the second contact surface **828** is in contact with the abutting member **824**. Specifically, the driving member **821** is rotatable relative to the stopper **822**, where the driving member **821** is formed with two cylindrical connection portions **829** protruding outward and disposed in the slide grooves **825** of the stopper **822**, and the driving force of the biasing member **823** causes the outer walls of the connection portions **829** to be always at least in contact with the inner walls of the slide grooves **825**.

(38) The shield assembly **80** includes at least one shield housing **83** rotatably connected to the main housing **22**. In this example, the shield assembly **80** includes one shield housing **83**. The shield housing **83** mates with the first rotating wheel housing **23** and the second rotating wheel housing **24** such that the first open space **231** and the second open space **241** can be substantially enclosed. The shield housing **83** includes a stopper portion **831** which can mate with the driving member **821**. When the shield assembly **80** is in the first state, the stopper portion **831** is in contact with the driving member **821** so that the shield assembly **80** and the housing assembly **20** remain relatively fixed to each other substantially. When the shield assembly **80** is in the second state, the stopper portion **831** is not in contact with the driving member **821**, and in this case, the shield housing **83** is rotatable relative to the main housing **22**. As another example, the shield housing and the main housing may be separate from each other, that is, the shield housing may be separated from the main housing when the shield assembly is in the second state. That is to say, the shield housing may be detached from the main body as a separately placed component. The preceding example may be understood as the case where when the locking assembly is at the open position, the user may remove the shield housing, or when the user does not want to mount the shield housing, the driving member is made to be at the open position and then the shield housing is placed separately, and the cutting can be continued. When the user wants to use the shield housing, the user only needs to mount the shield housing to the main body and drive the driving member to the closed position.

(39) As another possible example, the locking assembly **82** may be provided with no abutting member **824**, that is, the biasing member **823** directly abuts against the driving member **821**, and the preceding functions can also be implemented.

(40) The locking assemblies **82** are configured to be adjacent to the cutting region **26**. One group of locking assemblies **82** may be provided and used for locking the shield housing **83** and the main housing **22**. Two groups of locking assemblies **82** may be provided. In this example, the locking assemblies **82** are disposed on two sides of the cutting region **26**. Further, the two groups of locking assemblies **82** are configured to be symmetrical about the midplane **P1**. When the driving members **821** of the two groups of locking assemblies **82** are each at the open position, the shield housing **83** can be opened and is rotatable relative to the main housing **22**. It is to be noted that the two groups of locking assemblies **82** have substantially the same structure and are only disposed at different positions.

(41) The first rotating wheel housing **23** is provided with an elongated chip discharging port **232**. When the main body **10** runs, superfluous chips attached to the saw blade during the cutting are discharged via the chip discharging port **232**. The chip discharging port **232** is disposed at a position of the first rotating wheel housing **23** which is in contact with the shield housing **83** and adjacent to the cutting region **26**. The first rotating wheel housing **23** may be provided with multiple chip discharging ports **232**.

(42) As shown in FIG. 1, the handle assembly **60** further includes an auxiliary handle portion **66** for the user to hold. When using the main body **10**, the user may hold the handle portion **611** with one

hand and hold the auxiliary handle portion **66** with the other hand. This configuration facilitates the hold of the user. The auxiliary handle portion **66** is fixedly connected to the housing assembly **20**. Further, along the front and rear direction, the auxiliary handle portion **66** is disposed on the two sides of the midplane **P1** relative to the electric motor **31**, and along the left and right direction, the auxiliary handle portion **66** and the electric motor **31** are disposed on the left side of the handle midplane **P2**, which may be understood as the case where the handle portion **611** is not aligned with the auxiliary handle portion **66**, and the auxiliary handle portion **66** is fixedly connected to the battery pack mounting portion **27**, specifically, the auxiliary handle portion **66** is connected to the battery pack mounting portion **27** by screws. That is, the auxiliary handle portion **66** is disposed on a side surface of the battery pack mounting portion **27**. With this configuration, when the user controls the handheld band saw **100** with both hands, the user can very conveniently control the cutting direction of the handheld band saw **100**, thereby facilitating operation.

(43) The basic principles, main features, and advantages of the present application are shown and described above. It is to be understood by those skilled in the art that the preceding examples are not intended to limit the present application in any form, and the technical solutions obtained by means of equivalent substitution or equivalent transformation are all within the scope of the present application.

Claims

1. A handheld band saw, comprising: a housing assembly comprising a battery pack mounting portion for mounting a battery pack and a main handle formed with an accommodating space; a rotating wheel assembly at least partially disposed in the housing assembly and comprising a first rotating wheel and a second rotating wheel, wherein the first rotating wheel and the second rotating wheel are configured to be substantially symmetrical about a midplane; an electric motor disposed in the housing assembly and capable of driving one of the first rotating wheel and the second rotating wheel to rotate; a fan which is capable of generating an airflow in the housing assembly during rotation; and a control circuit board disposed in the accommodating space of the housing assembly, wherein the control circuit board is disposed on a side of the midplane facing the electric motor, the electric motor and the battery pack mounting portion are located on two sides of the midplane, the housing assembly is formed with an air inlet and an air outlet and, when the fan rotates, the airflow entering the housing assembly from the air inlet flows through at least the control circuit board and the electric motor.
2. The handheld band saw according to claim 1, wherein, when the fan rotates, the airflow entering the housing assembly from the air inlet flows through the control circuit board and then flows through the electric motor.
3. The handheld band saw according to claim 1, wherein the main handle is provided with the air inlet.
4. The handheld band saw according to claim 3, wherein the housing assembly comprises an electric motor housing formed with an interior space for accommodating the electric motor, the interior space communicates with the accommodating space, and the electric motor housing comprises a connection port from which the airflow enters the interior space.
5. The handheld band saw according to claim 4, wherein a flow guide structure disposed at an end of the main handle facing the electric motor is disposed in the accommodating space, the flow guide structure comprises a first guide portion and a second guide portion, and the first guide portion and the second guide portion cooperate with an inner wall of the main handle so that the airflow flows to the control circuit board and then flows to the connection port.
6. The handheld band saw according to claim 3, wherein the air inlet is disposed at a left end of the main handle, and the housing assembly is provided with the air outlet which is adjacent to the fan.
7. The handheld band saw according to claim 1, wherein the control circuit board is disposed in the

main handle and is configured to be adjacent to a left end of the main handle.

8. The handheld band saw according to claim 1, wherein the handheld band saw further comprises an auxiliary handle portion connected to the battery pack mounting portion.

9. The handheld band saw according to claim 8, wherein the main handle is configured to be substantially symmetrical about a handle midplane, and the auxiliary handle portion and the electric motor are located on a same side of the handle midplane.

10. The handheld band saw according to claim 1, wherein the air outlet and the air inlet are disposed on a same side of the midplane.

11. The handheld band saw according to claim 1, wherein the air outlet, the air inlet, and the control circuit board are disposed on one side of the midplane, and the battery pack mounting portion is located on another side of the midplane.

12. The handheld band saw according to claim 1, further comprising a shield assembly connected to the housing assembly and having at least a first state in which the shield assembly is fixed relative to the housing assembly and is capable of substantially covering the rotating wheel assembly and a second state in which the shield assembly is movable relative to the housing assembly, and a locking device connected to the housing assembly and comprising at least one group of locking assemblies, wherein each of the at least one group of locking assemblies comprises: a driving member having an open position which is retainable and a closed position which is retainable, wherein the shield assembly is in the second state when the driving member is at the open position, and the shield assembly is capable of being in the first state when the driving member is at the closed position; a stopper cooperating with the driving member so that the driving member is retainable at the open position or the closed position; and a biasing member providing a driving force for the driving member so that the driving member is always retained at a current position without being driven by an external force.

13. The handheld band saw according to claim 12, wherein the electric motor drives the first rotating wheel to rotate, the housing assembly comprises a first rotating wheel housing which substantially accommodates the first rotating wheel and a second rotating wheel housing which substantially accommodates the second rotating wheel, the shield assembly comprises a shield housing capable of mating with the first rotating wheel housing and the second rotating wheel housing to substantially cover the first rotating wheel and the second rotating wheel, and the first rotating wheel housing is provided with a chip discharging port.

14. A handheld band saw, comprising: a housing assembly comprising a battery pack mounting portion for mounting a battery pack; a rotating wheel assembly at least partially disposed in the housing assembly and comprising a first rotating wheel and a second rotating wheel, wherein the first rotating wheel and the second rotating wheel are configured to be substantially symmetrical about a midplane; an electric motor disposed in the housing assembly and capable of driving one of the first rotating wheel and the second rotating wheel to rotate; a control circuit board disposed in the housing assembly; and a fan which is capable of generating an airflow flowing through the control circuit board in the housing assembly during rotation, wherein the control circuit board and the electric motor are located on a same side of the midplane, wherein the housing assembly comprises a main handle formed with an accommodating space, the control circuit board is disposed in the accommodating space, and the main handle is provided with an air inlet, and wherein the air inlet is disposed at a left end of the main handle, and the housing assembly is provided with an air outlet which is adjacent to the fan.

15. The handheld band saw according to claim 14, wherein the airflow flows through the control circuit board and then flows to the electric motor.
